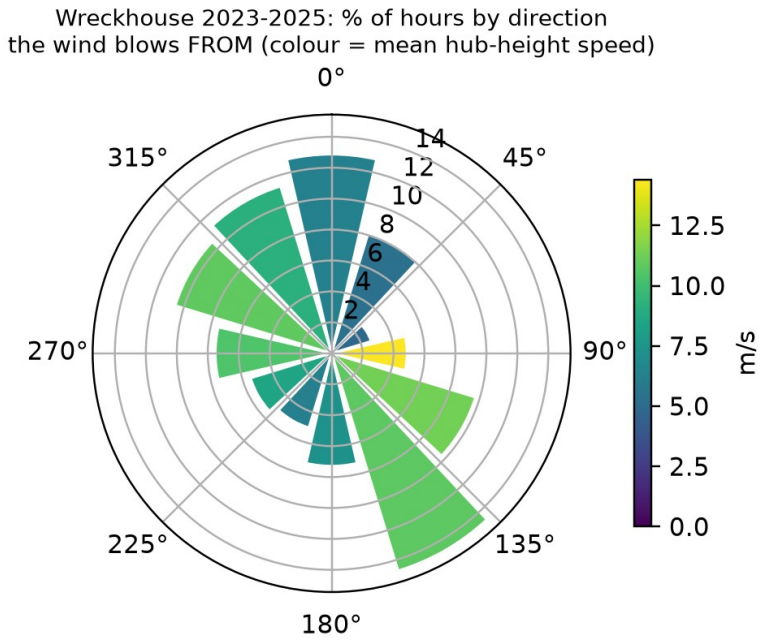
WakeWise NL

Where should wind turbines go on Newfoundland's west coast so they don't steal each other's wind? Real wind, real terrain, QAOA in Qiskit, and up to 24 qubits on IBM's quantum computer in Quebec

Team AI Entanglers

Mostafa Wasib · Mahmoud Al-Moghrabi

Qiskit Fall Fest 2026 · Memorial University



Real input: 26,264 hours of wind at Wreckhouse, NL, 2023 to 2025. Longer petals: directions the wind blows from more often.

WakeWise NL in one minute

154 GWh/yr

from the 9-turbine layout we recommend on our flat test parcel: the electricity of ~9,100 NL homes

~340x

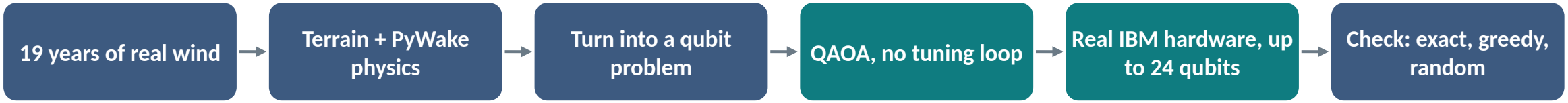
more often than random guessing, QAOA returns the best layout (flat parcel, 12 qubits, simulation)

24 qubits

real IBM hardware, real terrain: QAOA gets 66% of the way from random guessing to the best layout; the exact best appeared (chance: 1 in 17 million)

14 of 19 years

real wind years (2007-2025) in which the flat parcel's recommended layout is the best possible; a one-year design: 5 of 19



Teal = quantum steps. No quantum advantage is claimed: at this size we can check every answer exactly.

Strong NL wind meets a proposed hydrogen farm

Wreckhouse, southwest Newfoundland

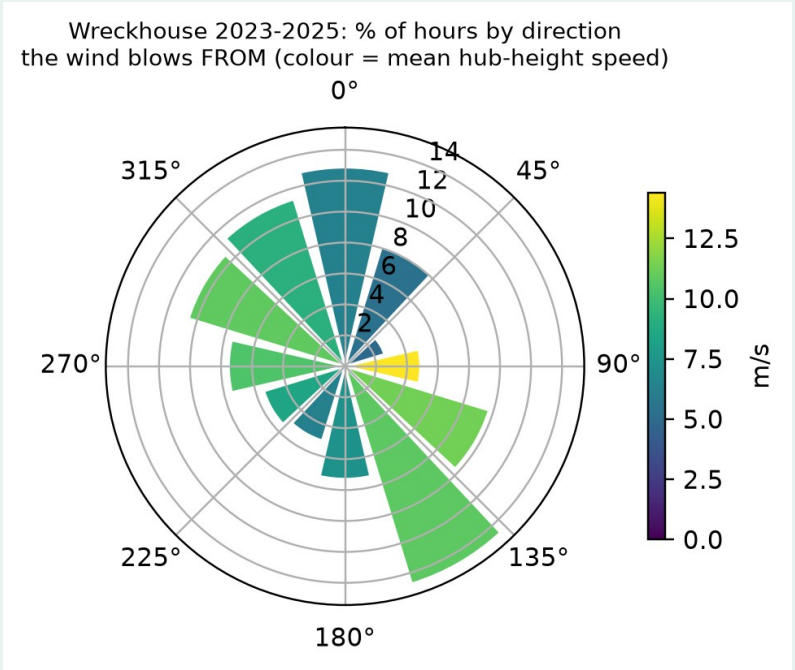
Gales here once blew trains off the tracks. Over 2023 to 2025 the wind at turbine height (110 m) averaged 9.0 m/s, so a 4 MW turbine would average 50% of its full output.

Project Nujio'qonik, next door

150+ turbines proposed on the Port au Port Peninsula and in the Codroy Valley to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

MUN's CREATE-SWIFT program asks for this

Compare classical and quantum methods on wind-energy optimization: accuracy, cost, scalability and real-world relevance.



The wind comes mostly from two opposite directions: south-east gales and north-westerlies. Turbines lined up along that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

Wakes waste wind, so siting is a real decision

17,600 MWh/yr

from one 4 MW turbine standing alone at Wreckhouse (2023-2025 wind, DTU PyWake)

up to 2.4%

lost to the wake (slower air) of one badly placed neighbour; a tightly packed block of 12 loses 4.2%

32% of hours

turbines are at full power anyway, so a wake costs almost nothing: wakes matter less here than at most sites

Decision and intended use

For each candidate site: build a turbine (1) or not (0). We report the best layouts, the energy at stake, and whether the choice holds up year to year. For early screening, not construction advice.

Sustainability benefit

More clean electricity, or green hydrogen, from the same land, steel and concrete; or the same energy from fewer turbines and less disturbed ground. The prototype shows the method; real benefit needs site-specific terrain, turbines and costs.

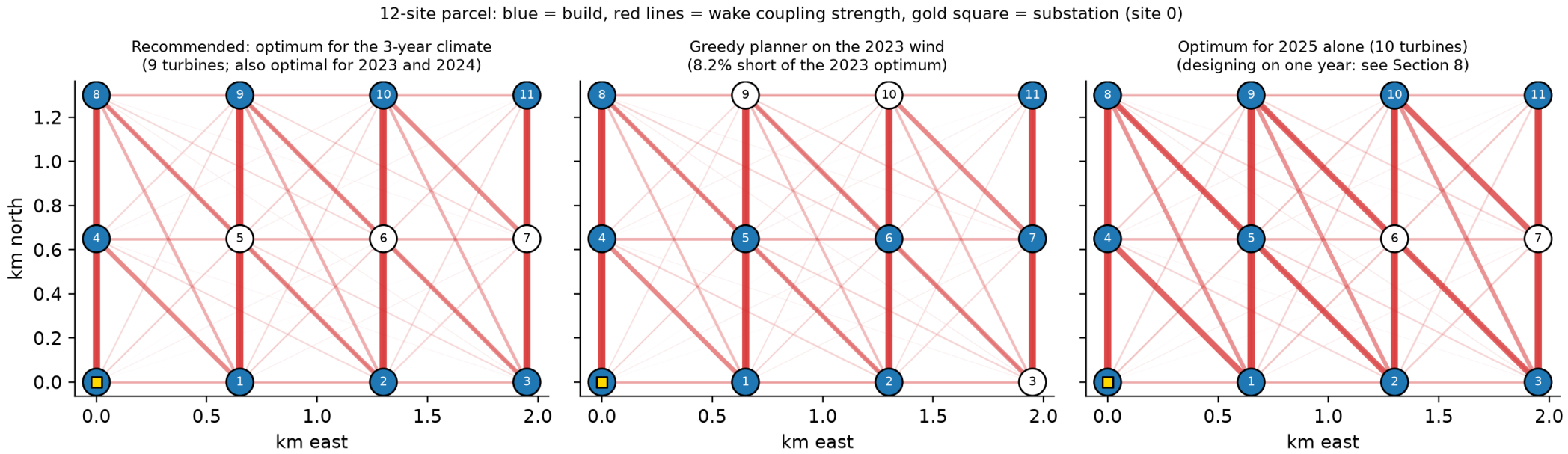
Inputs, output and scope

Input	What we used	Status
Wind	Wreckhouse weather station, hourly: 2023-2025 to design (26,264 h), 2007-2025 to test (163,068 h); scaled from 10 m up to the 110 m turbine height	**Measured (height scaling assumed)**
Wake physics	DTU PyWake, an industry-standard simulator; cross-checked with full-farm runs and a second wake model	**Industry-standard model**
Turbine	Generic 4 MW turbine with a 130 m rotor (PyWake's reference curves)	Illustrative
Costs	Each turbine's yearly cost = 88% of a lone turbine's energy, plus 2% per 650 m of cable; other levels tested	Illustrative, tested
Real-terrain parcel	Satellite elevation (Copernicus) and DTU's Global Wind Atlas; sites picked by a fixed rule	**Measured + modelled (screening)**
Output	One layout: build (1) or don't build (0) at each site	
Scope	Flat test parcel: 12-16 sites on a 650 m grid, 19 years, a wetland case. Real-terrain parcel: 12-24 sites. Simulation, noise model and real hardware	

Wakes make this a puzzle, not a checklist

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

In words: each turbine earns its energy E minus its cost c_i; each built pair loses W_{ij} to wakes (PyWake). A QUBO: yes/no choices with pair terms, in MWh/yr; lower is better



A simple greedy rule (keep adding the best next turbine) finds the 3-year optimum, but misses the best layout in 16 of 19 single years (up to 8.2% short). We check every answer against all 4,096 possible layouts.

Wake physics maps straight onto qubits

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i\, Z_i \;+\; \Sigma\, J_{ij}\, Z_i\, Z_j$$

Each site is a qubit. A turbine's value acts on its own qubit (a Z field); a wake loss links two qubits (a ZZ coupling). This is an Ising model.

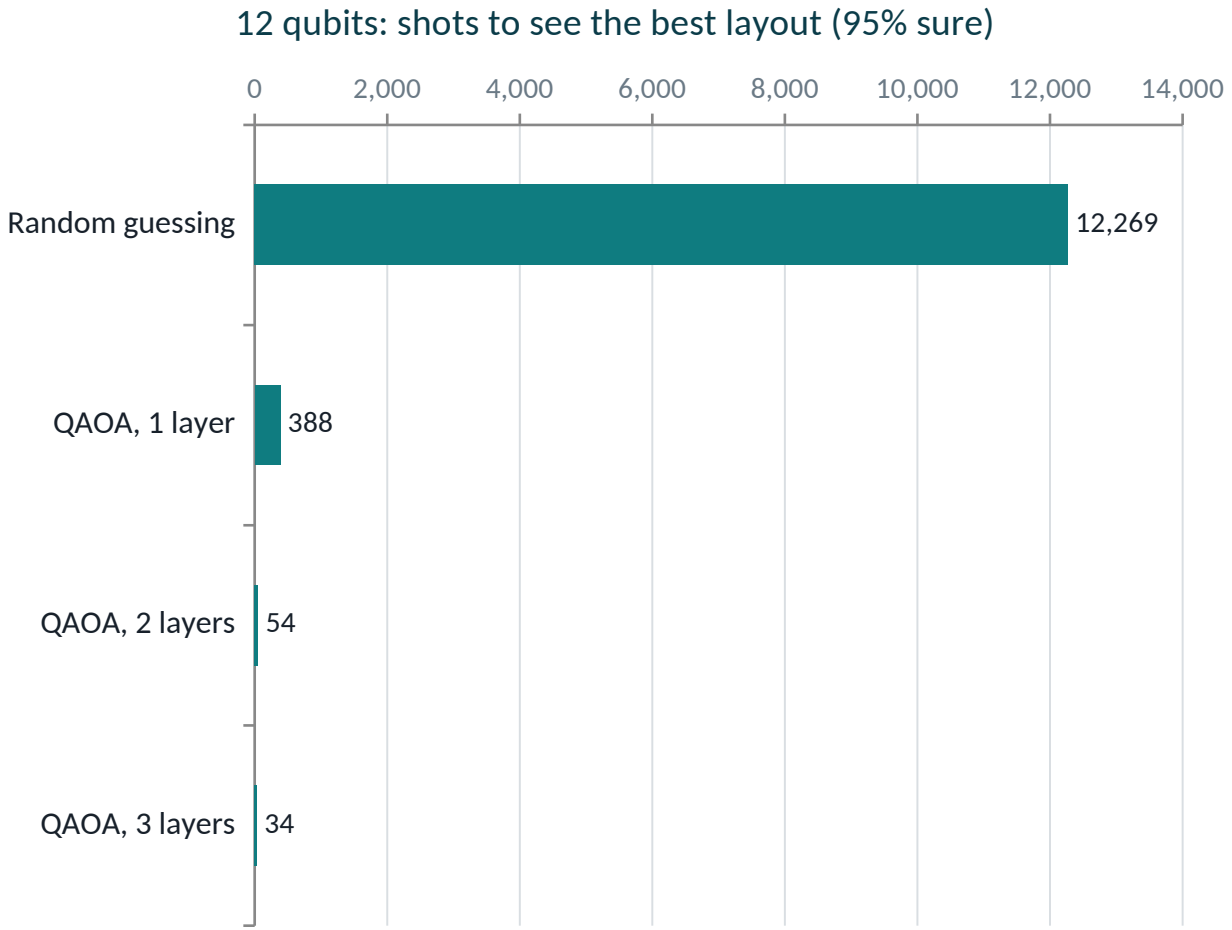
Why QAOA fits

- Already yes/no with pair terms: no extra qubits, no penalty tricks
- Grows fast: 24 sites give 16.8 million layouts (NP-hard in general)
- Tuned on the best 25% of samples (CVaR): a planner needs the best layout, not the average
- Small enough to check exactly, so we measure quality instead of assuming it

1. **Map** *classical*
   Wind and PyWake → QUBO → Qiskit QAOA circuit (qaoa_ansatz), 1 to 3 layers
2. **Optimize** *classical*
   Compile for IBM's chip; drop the weakest wake links to save two-qubit gates
3. **Execute** *quantum*
   Angles tuned in simulation, or predicted or reused at 20-24 qubits; run on ibm_quebec
4. **Post-process** *classical*
   Score every sampled layout; compare with exact, greedy and random

QAOA finds the best layout ~340x more often

Noiseless simulation



8.4% vs 0.02%

of shots give the best layout: QAOA with 3 layers vs random guessing

34 vs 12,269

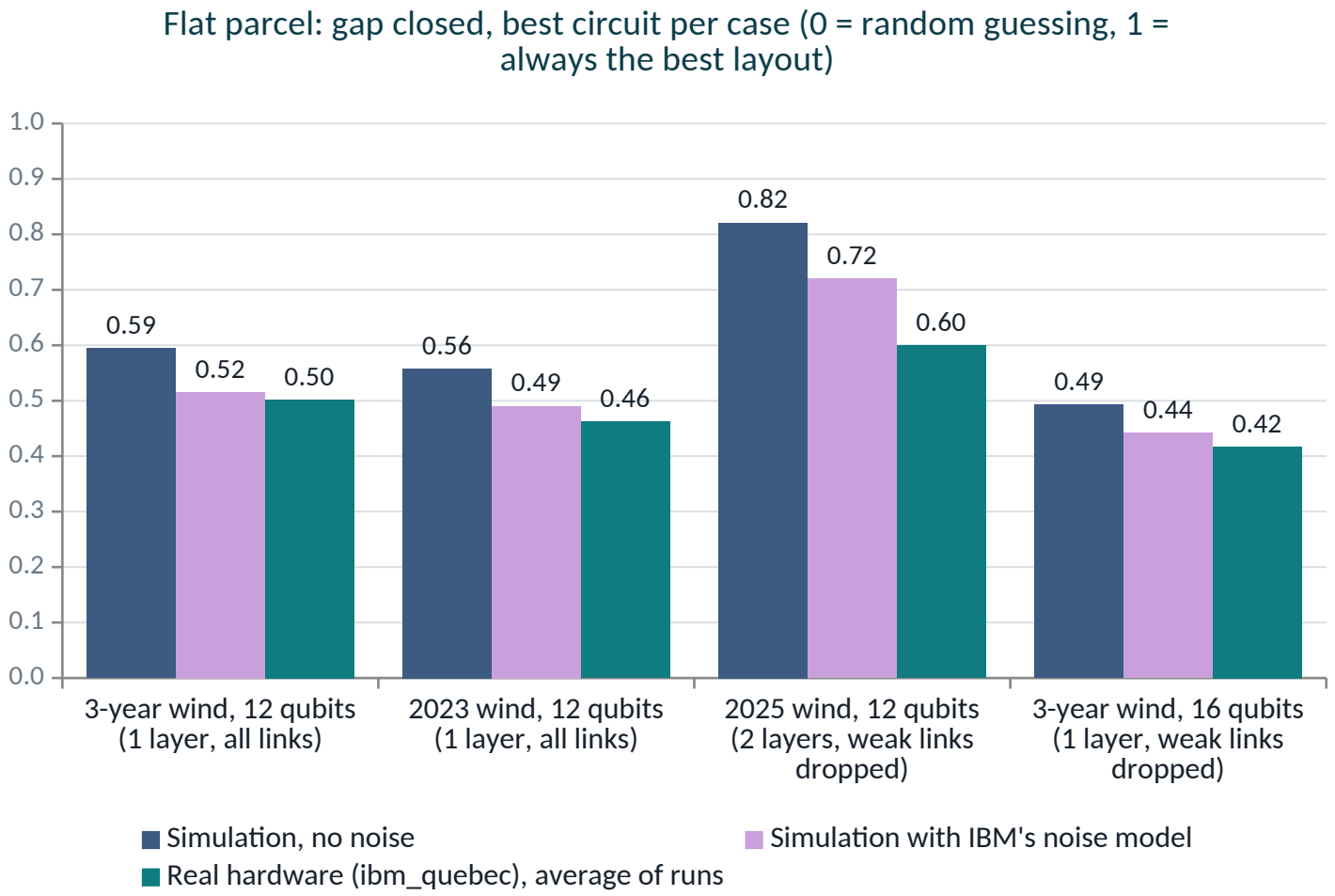
shots needed to be 95% sure of seeing it (one shot = one sampled layout)

20%

of QAOA samples (2 layers) beat the greedy rule in the 16 years where greedy misses

Honest caveat: a typical QAOA sample is good but not the best; the value is in its best few dozen shots.

On IBM's Quebec machine, the signal survives



Simulator + real hardware

ibm_quebec: IBM Quantum System One in Bromont, Quebec, 7 Nov 2026.

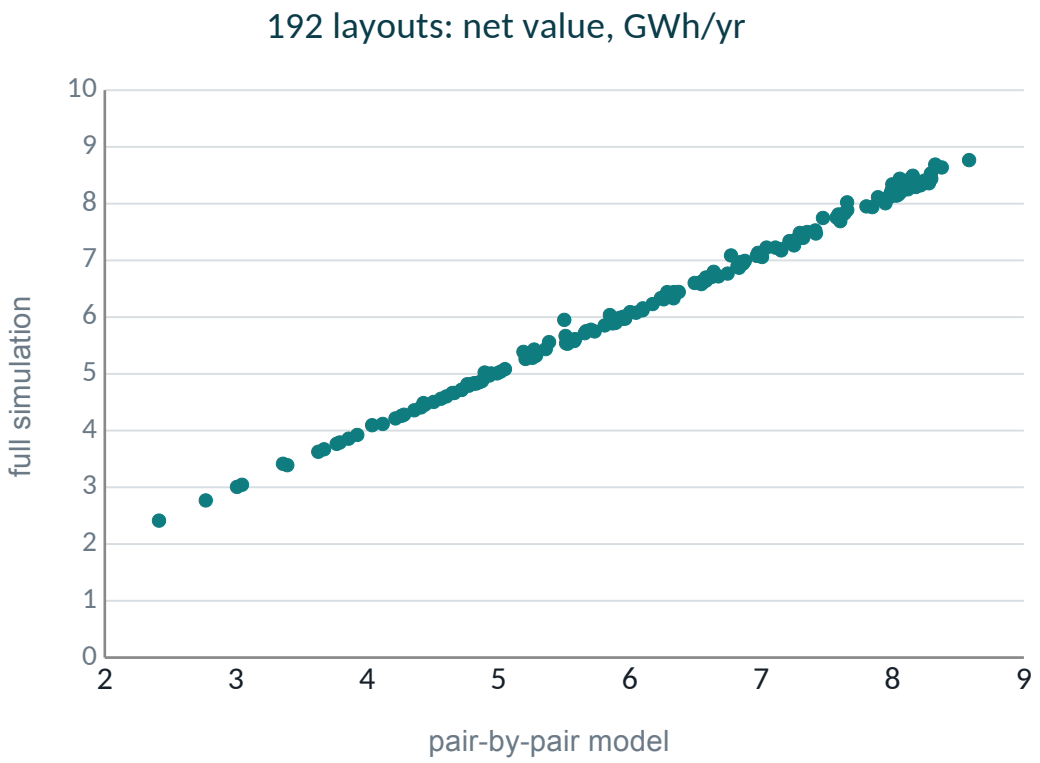
Main case, 4 runs: the best layout 23 times more often than random guessing.

2023 wind: 15% of hardware samples beat the greedy rule (random guessing: 2%).

A control circuit with no problem in it scores -0.02, the same as random guessing: the gain is QAOA, not noise.

We found a faulty qubit (101): it misread 1 as 0 in 65% of shots, though IBM's data looked normal. We now keep such qubits out of every run.

Validated: full simulations and 19 years of wind



Our pair-by-pair model ranks layouts in nearly the same order as full simulations (rank correlation 0.993; 1 = same order) and as a second wake model (0.992). Its best layout is the best in both.

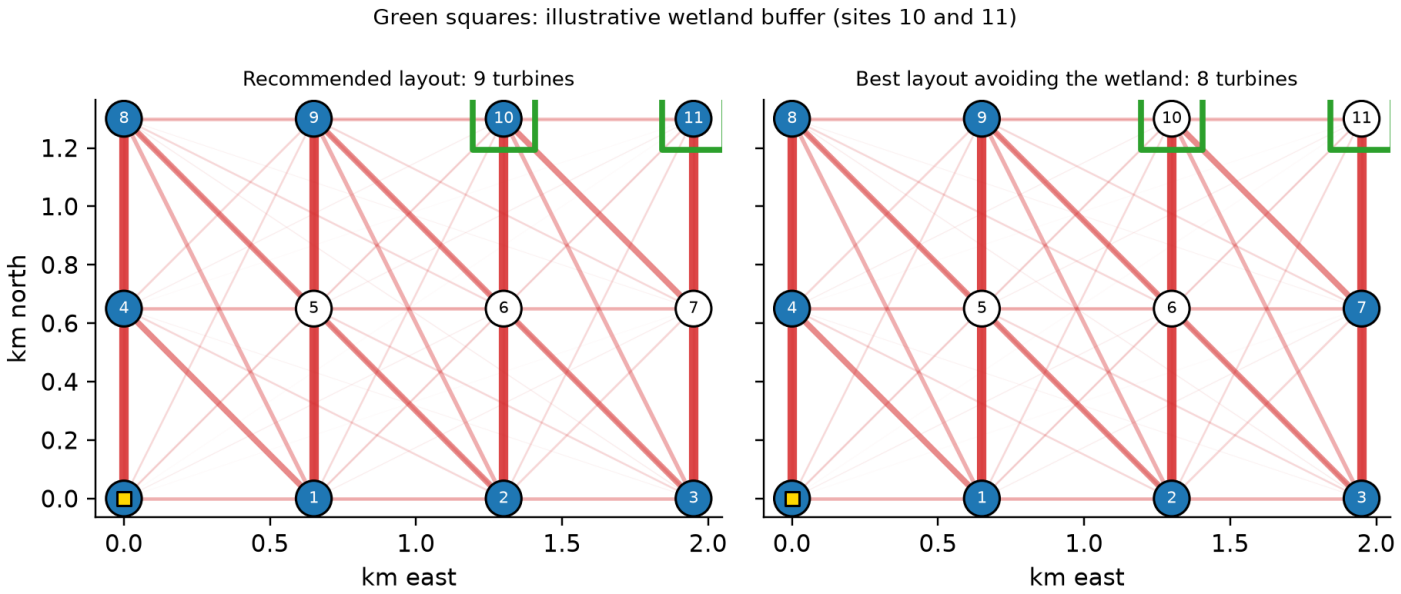
Layout designed on	Best in	Avg. shortfall	Worst year
2023-2025 (recommended)	**14 of 19**	**19 MWh/yr**	**167 MWh/yr**
2025 wind only	5 of 19	104 MWh/yr	416 MWh/yr

14 of 19 years

the recommended layout is the best possible; one designed on 2025 alone: 5 of 19

Wind changes from year to year, so we test each design on every year back to 2007. Shortfall = energy lost in a year compared with that year's best layout.

Protecting a wetland: what it costs



Illustrative scenario: two sites fall inside a wetland buffer. Next to Wreckhouse, the Grand Codroy Estuary is Newfoundland's only wetland of international importance (a Ramsar site).

519 MWh/yr

of net energy: the price of keeping turbines out of the wetland

9 → 8 turbines

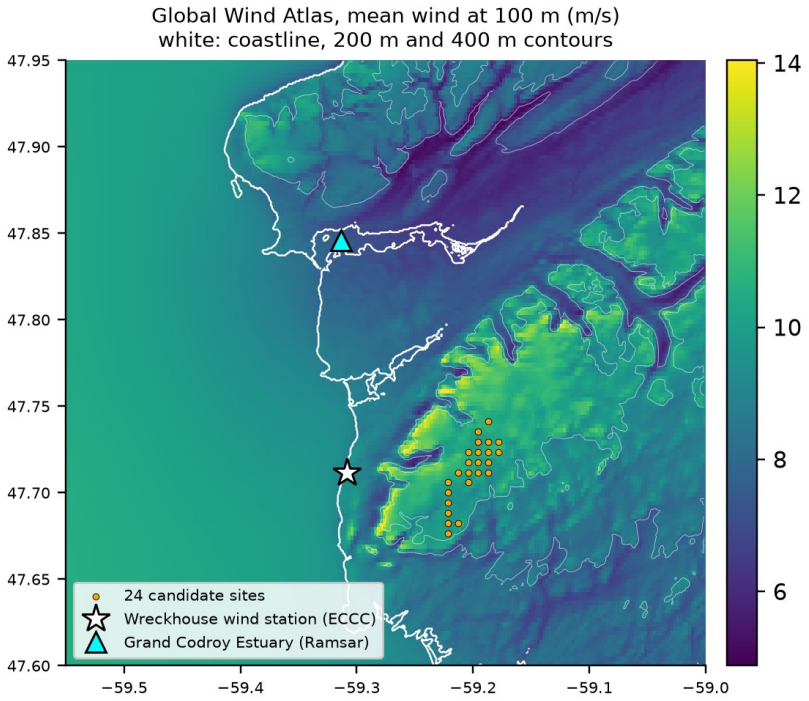
the optimizer re-plans the whole layout, not just drops two sites

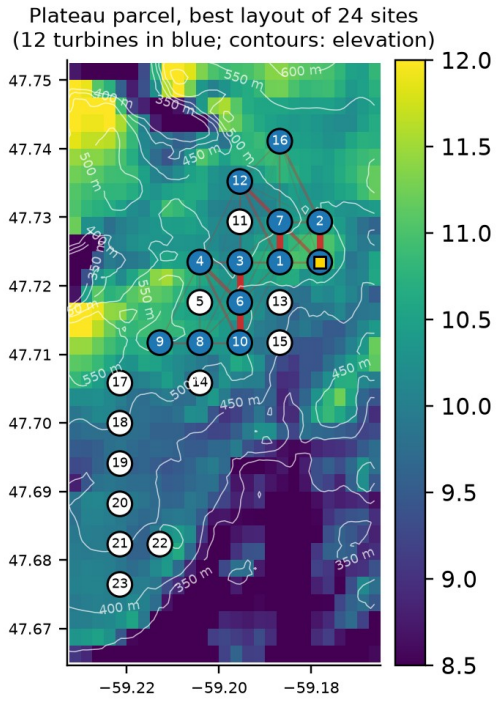
3.2% of shots

QAOA (3 layers, simulation) finds the new best layout, 133x random; the quantum program only adds a penalty on two qubits

A real parcel on the plateau above Wreckhouse

Screening level





Sites picked by a fixed rule: on land, slopes up to 10%, terrain like the weather station's (so the wind atlas can be trusted), 3 km from the Ramsar estuary, growing out from the windiest spot. Wind: the station's real hours, scaled per site by DTU's Global Wind Atlas.

423-550 m

above sea level: 24 buildable sites on the plateau, with 9.7-10.9 m/s average wind in the atlas

12 vs 4 years

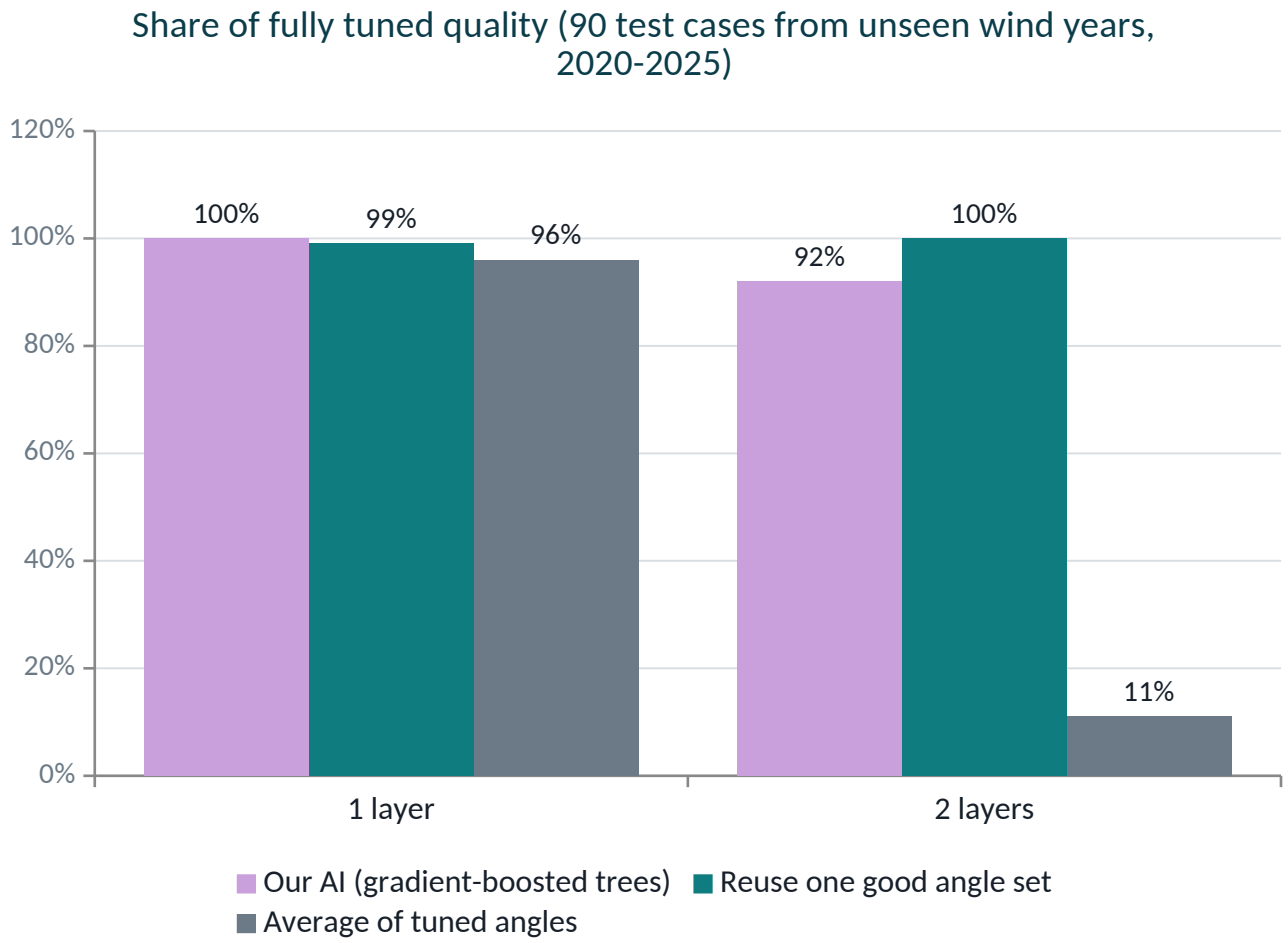
the layout is the best possible: designed on all 19 years vs on 2023-2025 only. On terrain, design on the long record

804x

more often than random guessing: QAOA (2 layers, simulation) finds the best layout for the first 16 sites

Our AI sets the angles; simple reuse works too

Simulation, unseen years



100% / 92%

our AI with 1 / 2 layers: no tuning loop needed

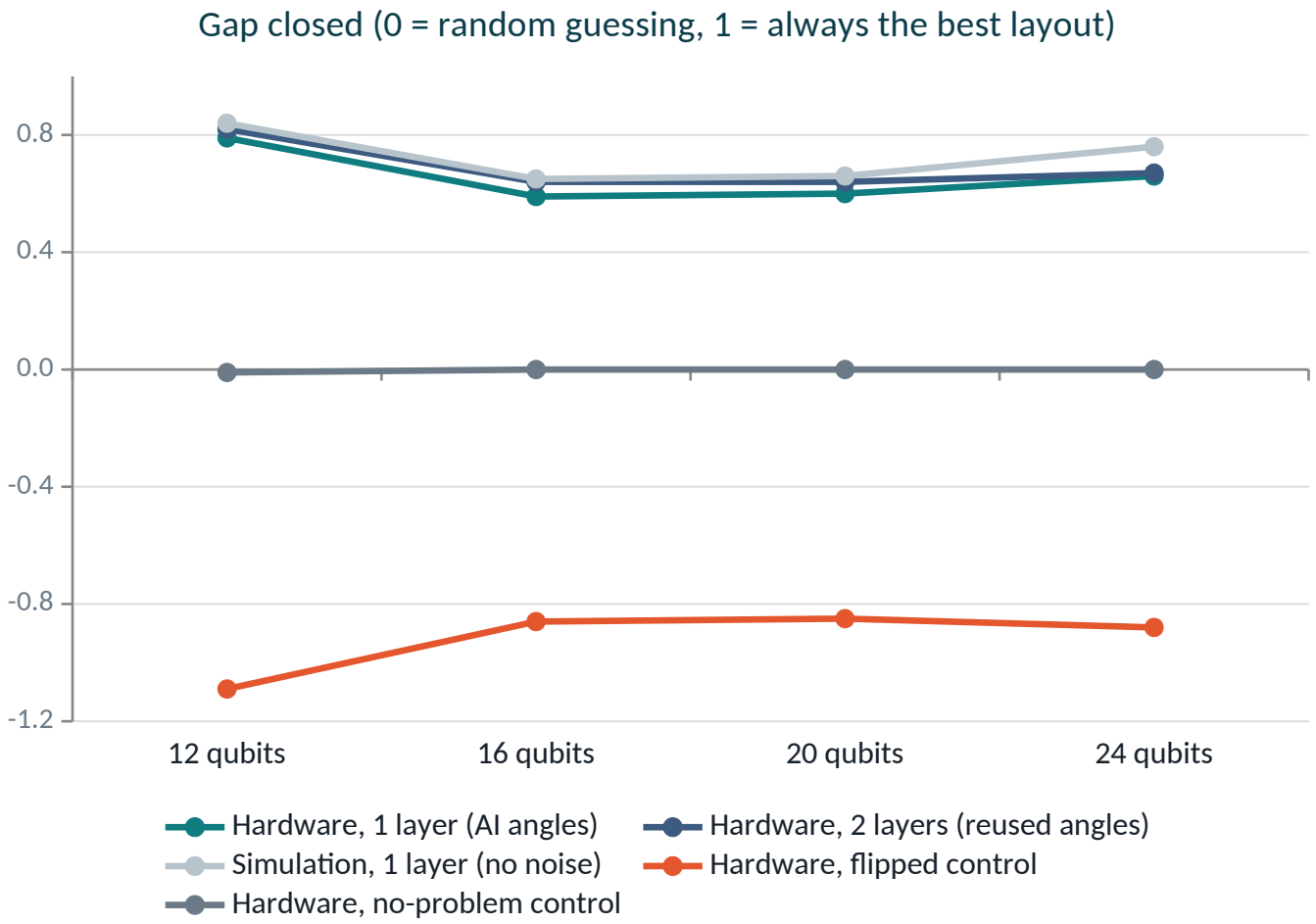
100%

reusing one good angle set (2 layers): beats our AI, an honest negative result

11%

averaging tuned angles (2 layers) fails: good angles differ, so their average lands in between

Real terrain: 24 qubits on IBM's Quebec machine



Real hardware

0.66 vs 0.76

gap closed at 24 qubits: real hardware vs noiseless simulation

4 in 24,000

shots gave the exact best of 17 million layouts; random guessing: 1 in 17 million

-0.88

flipped control: same gates, one sign reversed, so it steers toward bad layouts and lands below random. The machine follows the physics

What the evidence supports, and what it does not

Supports

- Siting on real NL wind is a real puzzle; our pair-by-pair model matches full simulations
- The recommended layout is the best possible in 14 of 19 real wind years
- QAOA favours the best layouts: ~340x random in simulation, and a clear signal on real hardware at 24 qubits
- No tuning loop needed at 20-24 qubits: our AI works, and reusing one angle set works as well
- Control circuits behave as physics predicts: the flipped circuit scores below random

Does not support

- Quantum advantage: a laptop still solves 24 sites exactly
- Bankable siting: the terrain parcel is a screening study, and costs are illustrative
- That our AI beats simply reusing one good angle set: it does not
- That the faulty qubit caused the weaker 2-layer results on the flat parcel: on healthy qubits they were no better

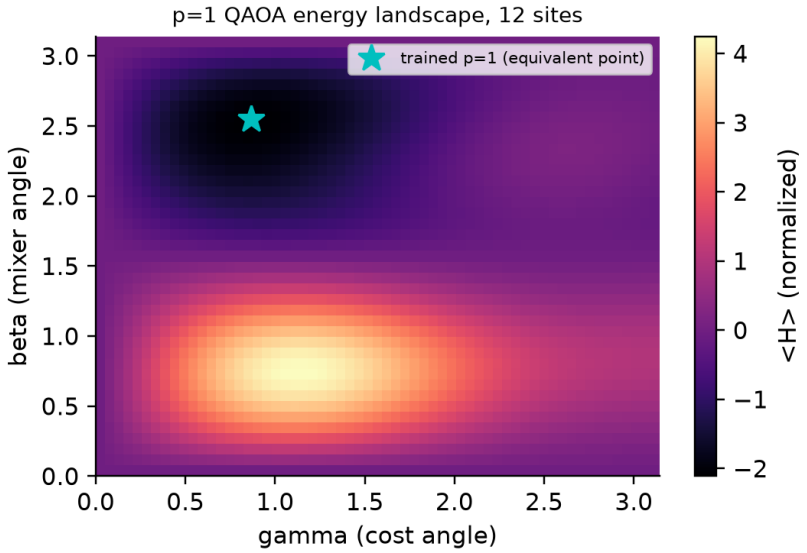
Next: measure wind on the plateau · add spacing and habitat limits · test bigger parcels against strong classical solvers (simulated annealing, MILP) · partner through MUN CREATE-SWIFT and Mitacs

Builds on Hancock et al. (2023) and Nigro et al. (2025), who set out quantum wind-farm layout and compare solvers. New here: a real NL site with 19 years of measured wind and real terrain, a check against full wind-farm simulations, and controlled hardware runs up to 24 qubits.

Resources: notebook about 30 minutes on a laptop; 21 hardware jobs, 174 s of quantum-computer time in total.

More clean energy from the same wind

- Real Wreckhouse wind, real terrain and DTU PyWake physics give a simple model that matches full wind-farm simulations
- The recommended layout is best in 14 of 19 real wind years; sparing a wetland costs about 519 MWh/yr
- QAOA in Qiskit finds it ~340x more often than chance in simulation, and keeps a clear signal on IBM's Quebec machine up to 24 qubits, with no tuning loop



QAOA's two angles for the 12-site parcel: darker means better layouts on average; the star marks the tuned angles.

github.com/mrw-soumik/Qiskit-AI-Entanglers-

Thank you. Questions?

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; IBM Qiskit 101 and the Fall Fest hardware example; DTU PyWake; ECCC data (Open Government Licence - Canada); DTU Global Wind Atlas (CC BY 4.0); Copernicus DEM (ESA); prior quantum wind-farm layout studies: Hancock et al. (2023), Nigro et al. (2025). Full credits in the notebook.